

Predicting Whether Institutions Have Above-Median Graduation Rates (IPEDS Analysis)

1. Data & Methodology

This analysis used IPEDS data including Graduation, Graduation Extended, Characteristics, Tuition and Costs, and Salaries datasets. Institutions with incoming cohorts of at least 200 students were included. A binary outcome variable was created based on whether an institution's graduation rate was above the median, rounded to three decimal places.

Multiple predictors were considered, including demographic composition, institutional characteristics, and financial variables.

2. Model Selection Process

Several modeling approaches were explored:

- LASSO logistic regression (penalized selection using AIC)
- Full logistic regression model
- Stepwise logistic regression

LASSO was selected as the final model due to its ability to reduce overfitting, handle multicollinearity, and produce a more interpretable set of predictors.

3. Final Model Results

The final LASSO model identified in-state tuition, gender composition, and selected demographic variables as the most important predictors of whether an institution's graduation rate exceeds the median.

Higher in-state tuition was positively associated with higher graduation rates, suggesting that institutional financial resources play a key role in student success.

Race-related variables showed strong associations, but these likely reflect broader structural differences across institutions rather than direct causal relationships.

4. Conclusion

The analysis shows that institutional resources and student composition are the primary drivers of graduation rate differences across institutions. LASSO provided a simplified and more stable model compared to traditional stepwise regression methods.